Quantum Mechanics II — Exercise Sheet 3

Boundary conditions and momentum

Variant: Solutions

Exercise 1 (A constant vector under twisted momentum). Let $\psi(x) = L^{-1/2}$ on $(0, L)$. For the self-adjoint momentum P_θ, with $0 < \theta < 2\pi$, find every momentum probability. (i) Decide whether $\psi \in D(P_\theta)$ and whether the second momentum moment is finite. (ii) Treat $\theta = 0$ separately.

Solution. The normalized eigenfunctions are $f_k^\theta = L^{-1/2} e^{\mathrm{i}(2\pi k+\theta)x/L}$, with momenta $\hbar(2\pi k + \theta)/L$. Direct integration gives

$$\langle f_k^\theta, \psi\rangle = \frac{1 - e^{-\mathrm{i}\theta}}{\mathrm{i}(2\pi k + \theta)}, \qquad p_k = \frac{4\sin^2(\theta/2)}{(2\pi k + \theta)^2}.$$

The boundary values of ψ are equal and nonzero, so they fail the twisted condition when $0 < \theta < 2\pi$. Moreover $p_k[\hbar(2\pi k+\theta)/L]^2$ is a positive constant independent of k. The second moment diverges, agreeing with the domain test. For $\theta = 0$ the vector is f_0^0, and the only outcome is momentum zero.

Exercise 2 (Neumann energy is a different observable). Consider $H_N = -\hbar^2 \mathrm{d}^2/(2m\mathrm{d}x^2)$ with Neumann conditions $\psi'(0) = \psi'(L) = 0$. You may take as given that this operator is self-adjoint and that the cosine basis is complete. (i) Find its normalized eigenfunctions and energies, including the lowest one. (ii) Compare its ground state with the Dirichlet box ground state.

Solution. Solving $-\psi'' = k^2\psi$ with both derivatives zero gives cosine modes. The normalized eigenfunctions and energies are

$$\upsilon_0 = L^{-1/2}, \quad E_0 = 0, \qquad \upsilon_n = \sqrt{2/L}\cos(n\pi x/L), \quad E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}, \quad n \geq 1.$$

The constant mode is admitted by the Neumann conditions and excluded by the Dirichlet conditions. The Dirichlet ground energy is positive, whereas the Neumann ground energy is zero. The differential expression alone therefore does not decide the observable.

Exercise 3 (A phase changes the boundary condition). Let $(G_\theta f)(x) = e^{\mathrm{i}\theta x/L} f(x)$ on $L^2(0, L)$. (i) Show that $G_\theta D(P_0) = D(P_\theta)$ and compute $G_\theta^{-1} P_\theta G_\theta$ on $D(P_0)$. (ii) Explain why this is not an equality of P_0 and P_θ on the same domain.

Solution. The multiplier has modulus one and preserves absolute continuity and square-integrability of the derivative. For a periodic f the endpoint values satisfy

$$(G_\theta f)(L) = e^{\mathrm{i}\theta}(G_\theta f)(0).$$

Multiplication by the inverse phase gives the converse inclusion of domains. Differentiating the product yields

$$G_\theta^{-1} P_\theta G_\theta = P_0 + \frac{\hbar\theta}{L}\mathbf{1} \quad \text{on } D(P_0).$$

The identity conjugates both action and domain. Unless θ is a multiple of 2π, the original domains differ.

Exercise 4 (Zero current does not choose a momentum). Let ψ be a real, smooth, normalized function with compact support inside $(0, L)$. (i) Show that it belongs to every $D(P_\theta)$ and that its probability current is zero. (ii) Explain why the momentum distributions for two distinct twists need not be the same. State a feature of their possible outcomes that distinguishes the observables.

Solution. Both endpoint values vanish, so every twisted boundary condition holds. The derivative is square-integrable. The current is

$$j(x) = \frac{\hbar}{m} \operatorname{Im}(\overline{\psi(x)}\psi'(x)) = 0.$$

Current is one local quantity and does not determine the full spectral distribution. For twists distinct modulo 2π, the two sets of possible momenta are disjoint:

$$\left\{ \frac{\hbar}{L}(2\pi k + \theta) : k \in \mathbb{Z} \right\}.$$

Since each distribution has total probability one on its own set, they cannot be the same measure on the momentum line.